the planets are made, many of these planetesimals are left over. The average time between civilization-threatening impacts on Earth is perhaps 200,000 years, twenty times the age of our civilization. Very different waiting times may pertain to extraterrestrial civilizations, if they exist, depending on such factors as the physical and chemical characteristics of the planet and its biosphere, the biological and social nature of the civilization, and of course the collision rate itself. Planets with higher atmospheric pressures will be protected against somewhat larger impactors, although the pressure cannot be much greater before greenhouse warming and other consequences make life improbable. If the gravity is much less than on Earth, impactors will make less energetic collisions and the hazard will be reduced—although it cannot be reduced very much before the atmosphere escapes to space.

The impact rate in other planetary systems is uncertain. Our system contains two major populations of small bodies that feed potential impactors into Earth-crossing orbits. Both the existence of the source populations and the mechanisms that maintain the collision rate depend on how worlds are distributed. For example, our Oort Cloud seems to have been populated by gravitational ejections of icy worldlets from the vicinity of Uranus and Neptune. If there are no planets that play the role of Uranus and Neptune in systems otherwise like our own, their Oort Clouds may be much more thinly populated. Stars in open and globular stellar clusters, stars in double or multiple systems, stars closer to the center of the Galaxy, stars experiencing more frequent encounters with Giant Molecular Clouds in interstellar space, may all experience higher impact fluxes at their terrestrial planets. The cometary flux might be hundreds or thousands of times more at the Earth had the planet Jupiter never formed—according to a calculation by George Wetherill of the Carnegie Institution of Washington. In systems without Jupiter-like planets, the gravitational shield against comets is down, and civilization-threatening impacts much more frequent.

To a certain extent, increased fluxes of interplanetary objects might increase the rate of evolution, as the mammals that flourished and diversified after the Cretaceous-Tertiary collision wiped out the